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School of Computing
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Tutorial Week 8

COMP3670 / COMP6670: Introduction to Machine Learning

Question 1 Gaussian elimination and free variables

This system has four equations in five variables:

$$\begin{aligned} x_1 + x_2 + x_4 + x_5 &= 2, \\ 2x_1 - x_3 + 3x_4 - 3x_5 &= 4, \\ x_2 + 2x_3 + x_4 + 4x_5 &= 3, \\ -4x_1 + 7x_2 + 13x_3 - 2x_4 + 31x_5 &= 7. \end{aligned}$$

(a) Write the augmented matrix and reduce it to reduced row-echelon form.

(b) Identify the pivot and free variables, then give every solution using two parameters.

Question 2 Checking whether a matrix has an inverse

Let

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -2 & 2 \\ 0 & 3 & 2 & -3 \\ 0 & 1 & 1 & -1 \\ 2 & -4 & -4 & 5 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & -2 & -2 & 2 \\ 0 & 3 & 2 & -3 \\ 0 & 1 & 1 & -1 \\ 1 & 5 & 3 & -5 \end{pmatrix}.$$

(a) Row-reduce $[\mathbf{A} \mid \mathbf{I}_4]$ to find $\mathbf{A}^{-1}$, then check your answer by multiplication.

(b) Write the fourth row of $\mathbf{B}$ as a combination of its first three rows. What does this tell you about invertibility? Confirm by giving a reduced row-echelon form of $\mathbf{B}$.

Question 3 One target, many coefficient vectors

Three vectors in $\mathbb{R}^4$ are

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 0 \end{pmatrix}.$$

Consider the targets $\mathbf{q} = (3, -1, 2, 7)^\mathsf{T}$ and $\mathbf{r} = (3, -1, 2, 8)^\mathsf{T}$.

(a) Verify that $\mathbf{w} = \mathbf{u} + 2\mathbf{v}$. What does this tell you about uniqueness of coefficients?

(b) Find every triple (α, β, γ) satisfying $\alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w} = \mathbf{q}$.

(c) Decide whether $\mathbf{r} \in \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$, and justify your answer using coordinates.

Question 4 Comparing column spaces

Suppose $\mathbf{A} \in \mathbb{R}^{m \times d}$ and $\mathbf{B} \in \mathbb{R}^{d \times k}$. Consider the two proposed inclusions

$$\text{Col}(\mathbf{AB}) \subseteq \text{Col}(\mathbf{A}) \qquad \text{and} \qquad \text{Col}(\mathbf{AB}) \subseteq \text{Col}(\mathbf{B}).$$

(a) Under what conditions on m and d does each comparison make sense as an inclusion of subspaces of the same ambient space?

(b) Prove that the first inclusion holds for every pair of compatible matrices.

(c) If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same size, both comparisons make sense. Must the second inclusion then hold? Give a counterexample if not.

Question 5 **Rank, nullity and pivot columns**

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 1 \\ -1 & -2 & 2 & -3 & 0 \\ 3 & 6 & -1 & 4 & 2 \\ 1 & 2 & 2 & -1 & 3 \end{pmatrix}.$$

(a) Reduce $\mathbf{A}$ to row-echelon form and identify the pivot columns of the original matrix.

(b) Use the rank–nullity theorem to find $\operatorname{rank}(\mathbf{A})$ and $\operatorname{nullity}(\mathbf{A})$.

(c) State the ambient spaces of $\operatorname{Col}(\mathbf{A})$ and $\operatorname{Nul}(\mathbf{A})$. Does $\operatorname{Col}(\mathbf{A})$ equal its ambient space? Explain.

Question 6 **Splitting a vector with a projection**

Let

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

(a) Verify that $\mathbf{P}^2 = \mathbf{P}$, then find $\operatorname{Col}(\mathbf{P})$ and $\operatorname{Nul}(\mathbf{P})$.

(b) More generally, suppose $\mathbf{A} \in \mathbb{R}^{d \times d}$ satisfies $\mathbf{A}^2 = \mathbf{A}$. Prove that every $\mathbf{x} \in \mathbb{R}^d$ can be written in exactly one way as $\mathbf{x} = \mathbf{u} + \mathbf{v}$, with $\mathbf{u} \in \operatorname{Col}(\mathbf{A})$ and $\mathbf{v} \in \operatorname{Nul}(\mathbf{A})$.

(c) Construct a 3×3 matrix $\mathbf{A}$ such that $\mathbf{A}^2 = \mathbf{A}$ and at least one entry of $\mathbf{A}$ is not 0, 1 or -1. Verify your example by multiplication.

Question 7 **When a power of a matrix vanishes**

Let $\mathbf{A} \in \mathbb{R}^{d \times d}$ and suppose $\mathbf{A}^q$ is the zero matrix for some integer $q > d$.

(a) Prove that $\mathbf{A}^d$ is already the zero matrix. Hint: if $\mathbf{A}^d\mathbf{x} \neq \mathbf{0}$, consider $\mathbf{x}, \mathbf{A}\mathbf{x}, \ldots, \mathbf{A}^r\mathbf{x}$, where r is the last exponent giving a nonzero vector.

(b) Give a 3×3 example for which $\mathbf{A}^3$ is zero but $\mathbf{A}^2$ is not. This shows that the exponent d cannot generally be reduced.